FORWARD DEPLOYED AI ENGINEERING

Capstone Project

Stage Two: Requirements

The product requirements document you write from your findings

Everything here must trace back to your discovery workbook. A requirement that cannot be traced is a preference, and preferences do not belong in a PRD.

WEEK ONE · DAYS FOUR AND FIVE

What this document is and why it matters

A product requirements document states what is being built, for whom, and how anyone will know when it is finished. It is not a design document and it is not a plan. It should be specific enough that two different engineers reading it would build broadly the same thing, and general enough that it does not dictate how they do it.

You will write version one at the end of week one and you will revise it during week two. That revision is compulsory. Write this version with the expectation that parts of it will turn out to be wrong, and record your assumptions clearly so that you can tell which ones failed when they do.

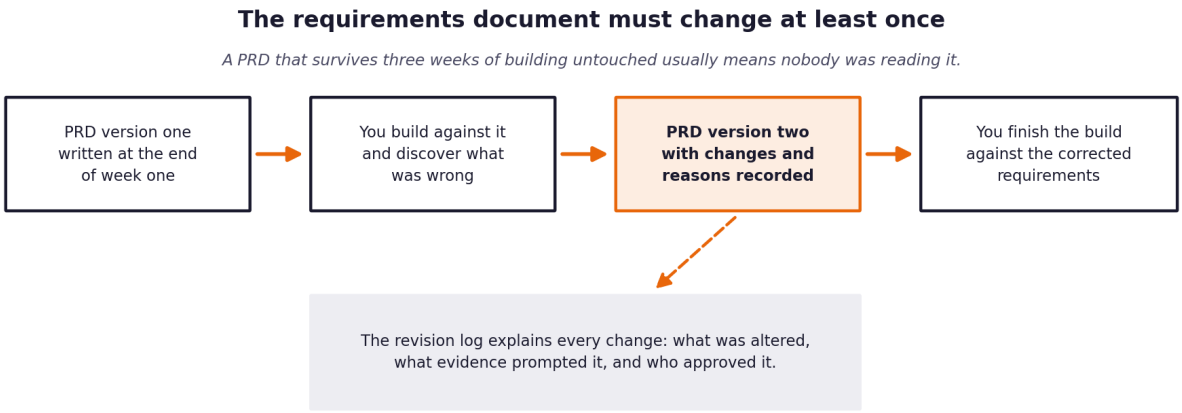


Figure 1. The requirements document is expected to change once building starts.

1 Document control

Field	Value
Version	1.0 — initial requirements reconstructed from capstone evidence; student review required
Written by	Kaushik Dutta
Date	2026-11-02 (submission preparation date; confirm actual authoring date)
Status	Draft for student review
Approved by	

2 The problem in one paragraph

Copy across the paragraph you wrote at the end of your discovery workbook. If you find yourself wanting to improve it as you copy, go back and improve it in the workbook first, so that the two do not drift apart.

Problem statement
CloudServe's support function faces slow replies, low FCR and poor satisfaction. A safe workflow should connect ticket triage to approved documentation, answer only when evidence and confidence support it, and otherwise hand off with context. Data and prototype limitations mean live business impact remains unproven.

3 Who this is for

Describe each group of users separately. A requirement that serves everybody usually serves nobody particularly well.

User group	What they need from this system	What they currently do instead	How you will know it worked for them
Customers raising tickets	Timely, accurate, transparent answer or human handoff	Wait, search docs, follow up	Live first reply, CSAT, FCR; disaggregate by group
Tier one support agents	Suggested intent/urgency, relevant	Search articles and personal snippets; triage	Review time, acceptance/edit rate,

	sources, safe draft and concise handoff	manually	safe escalation and workload
Tier two and specialist agents			
The head of support			

4 Functional requirements

Each requirement gets an identifier, a statement, a priority and a link back to the discovery evidence that produced it. The identifiers matter because your specifications, your prompts and your test cases will all refer to them.

Use the priority scale consistently. Must means the system is not acceptable without it. Should means it is expected but could be deferred with a stated consequence. Could means it would add value if time allows. Will not means it has been considered and deliberately excluded, and recording those is as useful as recording the rest.

ID	Requirement	Priority	Discovery evidence	Acceptance criteria
FR-01	Accept one ticket per request and preserve ticket ID/channel/tier/region/fluency.	Must	Stage 1 data source; interview flow	API validates required fields; record maps to source ticket.
FR-02	Predict intent with confidence and ranked alternatives.	Must	Marcus; labeled data	Return class, confidence, alternatives; evaluate per-class on labeled data.
FR-03	Predict urgency separately with confidence.	Must	Customer urgency; support SLA	Separate score; low confidence/high-risk routes to human; urgency metric reported.
FR-04	Retrieve relevant approved documentation chunks with IDs and scores.	Must	Ines; 29 article corpus	Return sources; retrieval evaluated against expected source IDs.
FR-05	Generate a cited draft only when gates pass and provider is configured.	Must	Ravi, Daniel	Every factual draft includes source IDs; provider failure safely escalates.
FR-06	Escalate uncertainty, high risk, unsupported	Must	Daniel, governance	Action=escalate/block; includes reason and sources

	and guarded cases with summary.			considered.
FR-07	Detect/block PII, injection, unsupported claims and tone/scope violations.	Must	Governance framework	Guardrail outcomes logged; adversarial and PII cases route safely.
FR-08	Persist decision trace for audit.	Must	Governance framework	SQLite record per processed ticket; no provider secret in record.
FR-09	Support unattended batch evaluation through input/output paths.	Must	Build spec; hidden 120 set	Clean command processes all records; produces metrics and per-ticket output.
FR-10	Support global auto-response kill switch.	Must	Governance framework	Presence of switch forces auto-response to human route.
FR-11	Provide human-readable health and API documentation.	Should	Submission/build spec	README documents setup, env, endpoints, evaluation.
FR-12	Allow Groq/OpenRouter optional generation with local fallback.	Should	User update; cost constraint	No required paid SDK; absent key safely degrades to escalation.

5 Non-functional requirements

These describe how well the system must do the things it does. They are frequently what distinguishes a demonstration from something that could actually be deployed.

ID	Category	Requirement	How it will be verified
NFR-01	Latency	p95 processing <3 seconds target; distinguish local pipeline from external model calls.	Harness percentiles; report measurement boundary.
NFR-02	Availability	Service degrades safely if provider, vector store or database fails.	Failure injection/manual checks; provider failure not yet live-tested.
NFR-03	Safety	Human-rated hallucination <=5% and citation accuracy >=95% before pilot.	Two assessors review >=50 drafts; not completed.
NFR-04	Calibration	Confidence calibration error <=5 percentage points.	ECE on held-out set; validation currently 16.6 points.
NFR-05	Fairness	Response quality gap <5pp across groups where sample supports analysis.	Dual-rated answer quality by subgroup; not available.
NFR-06	Privacy	No secrets or unnecessary PII in source or logs.	Credential/history scan; PII tests; inspect package.
NFR-07	Audit	Every decision stores action, reason, prediction, sources, guardrails and timing.	SQLite rows compared with processed tickets.
NFR-08	Reproducibility	Pinned dependencies and configurable paths; no hidden local paths.	Clean checkout run.
NFR-09	Security	Ticket text treated as untrusted; provider keys loaded from	Injection cases; config review.

		environment.	
NFR-10	Maintainability	Prompts, evaluation and governance artifacts versioned.	Repository review.

6 What is deliberately out of scope

Recording what you are not building is one of the most valuable things a requirements document does, because it is the section that prevents an argument in week three.

Not building	Why not	What would have to change for this to be reconsidered
Autonomous account or security changes	High-impact actions require human ownership	Formal authorization, security review and rollback design.
Replace support agents	Risk of unsafe or inequitable service	No evidence supports fully autonomous support.
Train a new foundation model	Out of scope/cost and data rights	Validated data, budget and explicit requirement.
Customer-facing production deployment	Prototype evidence is insufficient	Meet safety/calibration/fairness and live-pilot gates.
Claim SLA/CSAT improvement from offline timing	No live event data	Instrument production baseline and controlled pilot.

7 Assumptions and their consequences

Every assumption you are making is a place where this document could turn out to be wrong. Listing them now is what makes the revision in stage five a considered update rather than a scramble.

Assumption	Why you believe it	What happens if it is false	How you will find out
Docs have relevant coverage for many cases	71.4% dev label answerability	Retrieval/automation opportunity smaller	Compare judged answers with source docs and human audit.
Ticket labels are reliable enough for offline evaluation	Pack supplies labels	Metrics may reflect synthetic conventions	Audit sample and compare with subject-matter review.
Human agents can review suggested handoffs	Interview requests summaries/sources	Workflow could increase rather than reduce effort	Pilot measure adoption/review minutes.
Students can access a provider free tier	Project pack suggests options	Generation unavailable at demo	Demonstrate safe local escalation fallback.
Historical FCR/CSAT proxy current service	Brief narrates similar figures	Baseline comparison misleading	Collect live timestamped outcomes.

8 Success measures

State the measures this system will be judged on, with the baseline and the target for each, and say who is accountable for reporting them.

Measure	Baseline	Target	Measured how	Reported by
First contact resolution	43.8% history (main brief says ~42%)	>=60%	Resolved first contact / tickets in live pilot	Support lead
Time to first reply	Not in data; narrative says 8–12h	<5 minutes system target; contractual SLA 2h	Ticket created to first human/customer-visible response	Support ops
CSAT	2.972/5 history	>=4.0/5	Post-contact rating	Support lead
Escalation	37.8% expected	<=30%	Human handoff /	Support ops

	route (labels)		valid tickets	
Urgency accuracy	43.75% validation	Improve to agreed safe threshold; no pass claim	Labeled held-out evaluation + safety audit	Engineering
Citation accuracy	53.3% automated proxy; human unknown	>=95% human-rated	Two independent assessors on >=50 responses	Governance owner
Hallucination	46.7% automated grounding block proxy; human unknown	<=5% human-rated	Two independent assessors on >=50 responses	Governance owner

9 Open questions

The things you do not yet know. Each one needs an owner and a date by which it will be resolved, otherwise it will still be open in week three.

Question	Why it matters	Owner	Resolve by
What are actual first reply times by tier/channel?	SLA and subgroup experience	Support operations	Before any impact claim
Which prompt/provider/model version is approved?	Reproducibility and security	Engineering owner	Before pilot
Who are two independent answer assessors?	Required human quality review	Governance owner	Before pilot
What are retention/deletion/access policies?	Privacy and audit safety	Security/privacy owner	Before live data
How much personal sprint capacity was available?	Honest planning/effort record	Kaushik Dutta	Complete from records

The review before you move on

Read your finished PRD and ask, for each requirement in turn, whether an engineer who had not attended any of your interviews could build it from what is written. Where the answer is no, the requirement is not finished. This check takes twenty minutes and it is the single highest-value thing you can do at the end of week one.